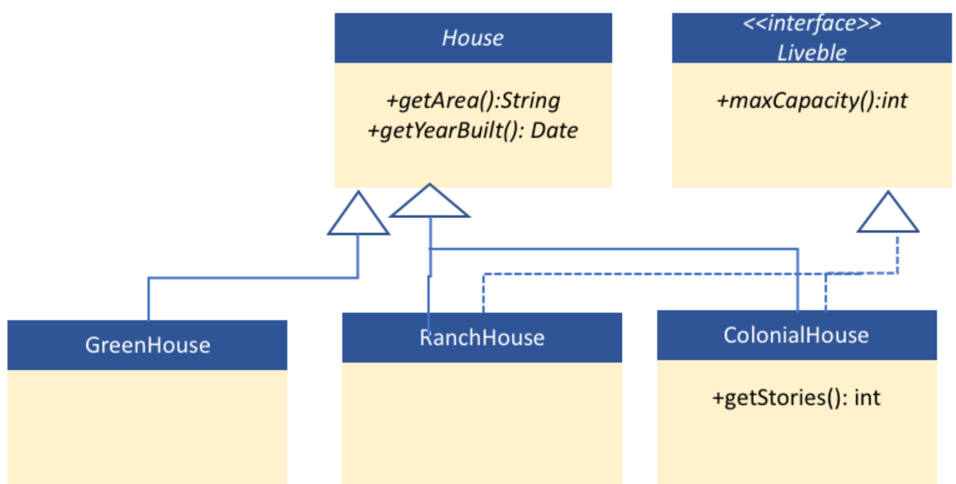
New York University
Computer Science Department
Introduction to Software Development
Dr. Nader Nassar
Homework 9
Deadline: *See NYUclasses for the deadline, 5% off per day after the deadline (3 days maximum).*

Please read the guidelines carefully to avoid receiving a zero on the HW:

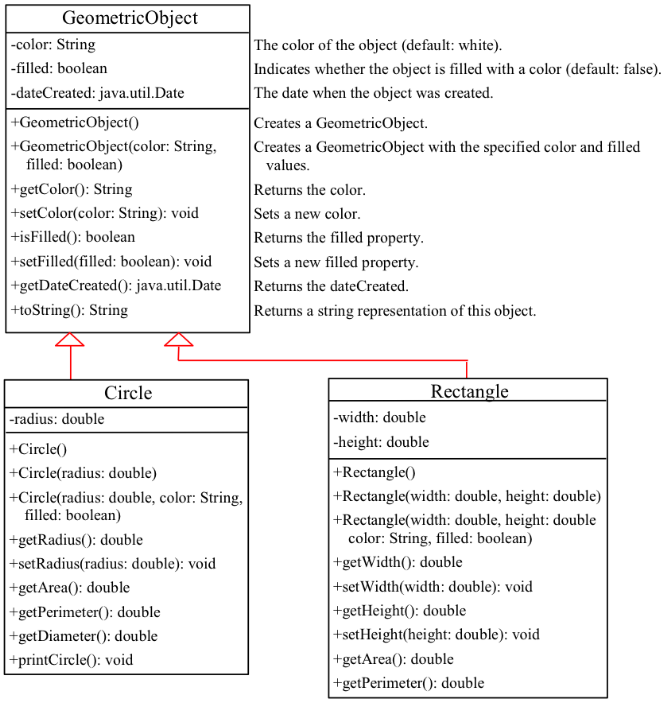
- Create a folder for each exercise and include all your java classes in the same folder. Name this folder ExerciseN where N is the exercise number
- Apply this convention to all future homework assignments during the semester.
- Compile and run the program for each exercises.
- Make an archive (zip file or compressed file) with all the java files (the .java files NOT the .class files) and post it on NYU Classes (StudentName_Homework_X.zip)
- It is your responsibility to make sure that the Zip files have your actual files. You may send the file to yourself by email to double check before you upload on NYU classes.
- If the graders cannot open the file, you will receive a grade of zero.
- If you send the .class files instead of the .java files (source files) you will
- receive a zero.
- Cheating will be severely addressed with an immediate zero on the homework and a report to the academic advisor and the administration.
- You will automatically lose 50% of the points for an exercise if the program does not compile and run correctly

Question1 [20 points]
Implement the following UML diagram in Java. Create all necessary classes and interfaces



Question 2 [30 points]

A. Implement the following UML diagram in Java



B. Create two custom exceptions one used by Circle and the other used by both Circle & Rectangle classes
   i. InvalidRadiusException is thrown by Circle class if the Radius is < 0
      See "*Custom Exception Class Example*" slide in the lecture for reference
   ii. InvalidColorException is thrown by both Circle & Rectangular class if the filling color is set to "Red" that return a text message saying " color can't be Red"

C. Create a test class (with main method) named TestObjectsWithException which does the following
   i. Create three Circle objects, one with radius:-1 , Color Red , second with Radius: 5, Color Red, and a third with Radius -5 and Color black
   ii. Create two Rectangular objects, one with width & height4,5 color: Green , a second with width & height: 3,5 color: Red.

Question 3 [30 points]

A. Write an Octagon class that inherits from GeometricObject and implements the Comparable interface, comparing the area of the shape. You can assume that all the sides are equal in length.

B. Implement the Comparable interface in the Circle and Rectangle from Question2 comparing the area of each shape. *(reuse the same classes in Question 2)*

C. Also add the overloaded toString() method to each class that prints a summary of the object. Then, in another class, create an ArrayList of at least one of each Circle, Rectangle and Octagon .

D. Write 2 methods:

i. A sumArea() method that takes an ArrayList of GeometricObjects and returns the sum for the entire list

ii. A printSorted() method that takes an ArrayList, sorts it by area (using the compareTo() method) and then prints the sorted list using the overloaded toString() method.

Question4 [10 points]

Create a java program that reads the file NameList.txt and returns the count of the names Bob & Alice

The output of the program should be as follows:

There are N Bob and M Alice in file NameList.txt

Where N & M are the actual count of each word

Question5 [10 points]

Create a new class and name it TimeDelayRecusion that does exactly what TimeDelay class, below, does. <u>You will have to redesign TimeDelay method to use Recursion instead of a loop.</u>

Please submit both classes, TimeDelay.java class and TimeDelayRecusion.java

```java
public class TimeDelay {

    public TimeDelay(int n) {
        while(n > 0) {
            System.out.println(n + "...");
            oneSecondDelay ();
            n -= 1;
        }

    }

    public void oneSecondDelay() {
        try {
            Thread.sleep(1000);
        }
        catch (InterruptedException ignore) {
        }
    }

    public static void main(String[] args) {
        TimeDelay c = new TimeDelay(10);

    }
}
```